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Code JupyterLab Python 3 (ipykernel)

[1]:
```python
print("----- LIST EXAMPLE -----")

# create list (duplicates allowed)
fruits = ["apple", "banana", "cherry", "apple"]

# print original list
print("Original List:", fruits)

# Add element
fruits.append("orange")
print("After adding orange:", fruits)

# Remove element
if "banana" in fruits:
    fruits.remove("banana")
    print("After removing banana:", fruits)

# Access element
print("First element:", fruits[0])

# Loop through list
print("All elements:")
for item in fruits:
    print(item)
```

```
----- LIST EXAMPLE -----
Original List: ['apple', 'banana', 'cherry', 'apple']
After adding orange: ['apple', 'banana', 'cherry', 'apple', 'orange']
After removing banana: ['apple', 'cherry', 'apple', 'orange']
First element: apple
All elements:
apple
cherry
apple
orange
```

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Code

JupyterLab Python 3 (ipykernel)

[2]:
```python
print("----- TUPLE EXAMPLE -----")

# create tuple
numbers = (10, 5, 20, 3, 10, 5)

# print tuple
print("Tuple:", numbers)

# access element
print("First element:", numbers[0])

# loop through tuple
print("All elements:")
for num in numbers:
    print(num)

# count and index
print("Count of 10:", numbers.count(10))
print("Index of 20:", numbers.index(20))
```

```
----- TUPLE EXAMPLE -----
Tuple: (10, 5, 20, 3, 10, 5)
First element: 10
All elements:
10
5
20
3
10
5
Count of 10: 2
Index of 20: 2
```

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Code JupyterLab Python 3 (ipykernel)

[3]:
```python
print("----- TUPLE PROGRAM -----")

# create tuple with float values
numbers = (10.5, 20.3, 10.5)

# print tuple
print("Tuple:", numbers)

# access using index
print("Element at index 0 (first):", numbers[0])
print("Element at index -1 (last):", numbers[-1])

# loop through tuple
print("\nAll elements:")
for num in numbers:
    print(num)

# count and index
print("\nCount of 10.5:", numbers.count(10.5))
print("Index of 20.3:", numbers.index(20.3))
```

```
----- TUPLE PROGRAM -----
Tuple: (10.5, 20.3, 10.5)
Element at index 0 (first): 10.5
Element at index -1 (last): 10.5

All elements:
10.5
20.3
10.5
```

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Code JupyterLab Python 3 (ipykernel)

[4]:
```python
print("----- TUPLE UNPACKING -----")

# create tuple
numbers = (10.5, 20.3, 10.5)

# unpack tuple into variables
a, b, c = numbers

# print values
print("Tuple:", numbers)
print("a =", a)
print("b =", b)
print("c =", c)
```

```
----- TUPLE UNPACKING -----
Tuple: (10.5, 20.3, 10.5)
a = 10.5
b = 20.3
c = 10.5
```

[5]:
```python
# create a list of 5 numbers and print all elements
print("----- LIST PROGRAM -----")

# create list
numbers = [1, 2, 3, 4, 5]

# print all elements
print("List elements:")
for num in numbers:
    print(num)
```

```
----- LIST PROGRAM -----
List elements:
1
2
3
4
5
```

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Code JupyterLab Python 3 (ipykernel)

[6]:
```python
# print first and last element of a list [ 10 , 20 , 30 , 40 , 50 ]
print("----- LIST PROGRAM -----")

# create list
numbers = [10, 20, 30, 40, 50]

# print list
print("List:", numbers)

# first element
print("First element:", numbers[0])

# last element
print("Last element:", numbers[-1])
```

```
----- LIST PROGRAM -----
List: [10, 20, 30, 40, 50]
First element: 10
Last element: 50
```

[7]:
```python
# add number 100 to the list [1,2,3,4]
print("----- ADD ELEMENT -----")

# create list
numbers = [1, 2, 3, 4]

# add 100 to list
numbers.append(100)

# print updated list
print("Updated List:", numbers)
```

```
----- ADD ELEMENT -----
Updated List: [1, 2, 3, 4, 100]
```

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Code JupyterLab Python 3 (ipykernel)

[8]:
```python
# remove element 3 from the list[1,2,3,4,5]
print("----- REMOVE ELEMENT -----")

# create list
numbers = [1, 2, 3, 4, 5]

# print original list
print("Original List:", numbers)

# remove element 3
numbers.remove(3)

# print updated list
print("After removing 3:", numbers)
```

```
----- REMOVE ELEMENT -----
Original List: [1, 2, 3, 4, 5]
After removing 3: [1, 2, 4, 5]
```

[9]:
```python
# find the sum of all elements in [5,10,15,20]
print("----- SUM OF ELEMENTS -----")

# create list
numbers = [5, 10, 15, 20]

# calculate sum
total = sum(numbers)

# print result
print("List:", numbers)
print("Sum:", total)
```

```
----- SUM OF ELEMENTS -----
List: [5, 10, 15, 20]
Sum: 50
```

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Code

```
Sum: 50
```

[10]:
```python
#find the maximum and minimum in [8,3,9,1,6]
print("----- MAX & MIN -----")

# create list
numbers = [8, 3, 9, 1, 6]

# find max and min
maximum = max(numbers)
minimum = min(numbers)

# print results
print("List:", numbers)
print("Maximum:", maximum)
print("Minimum:", minimum)
```

```
----- MAX & MIN -----
List: [8, 3, 9, 1, 6]
Maximum: 9
Minimum: 1
```